$$y_h' = c_1 e^t(\cos 2t - 2\sin 2t) + c_2 e^t(\sin 2t + 2\cos 2t)$$

from which we obtain the system

$$0 = c_1$$

$$2 = c_1 + 2c_2.$$

Hence $c_1 = 0$ and $c_2 = 1$ and a solution to the ODE is given by

$$y(t) = \int_0^t (1/2)e^{t-v}\sin(t-v)g(v)\ dv + e^t \sin 2t.$$

SECTION 7.8

1. $\int_{-\infty}^{\infty} (t^2 - 1)\delta(t)\ dt$

To evaluate this integral we use the fact that

$$\int_{-\infty}^{\infty} f(t)\delta(t)\ dt = f(0).$$

Therefore

$$\int_{-\infty}^{\infty} (t^2 - 1)\delta(t)\ dt = (0^2 - 1) = -1.$$

3. $\int_{-\infty}^{\infty} (\sin 3t)\delta\left(t - \frac{\pi}{2}\right)\ dt$

To evaluate this integral we use the fact that

$$\int_{-\infty}^{\infty} f(t)\delta(t-a)\ dt = f(a).$$

Therefore

$$\int_{-\infty}^{\infty} (\sin 3t)\delta\left(t - \frac{\pi}{2}\right)\ dt = \sin 3\left(\frac{\pi}{2}\right) = -1.$$

5. $\int_0^\infty \; e^{-2t}\delta(t-1)\; dt$

To evaluate this integral we use the fact that $\int_b^c \; f(t)\delta(t-a)\; dt$ is equal to $f(a)$ if $a \in (b,c)$ and is equal to zero if $a \notin (b,c)$. Therefore

$$\int_0^\infty \; e^{-2t}\delta(t-1)\; dt = e^{-2(1)} = e^{-2}.$$

7. $\delta(t-1) - \delta(t-3)$

By definition, the Laplace transform of this function is given by

$$L\{\delta(t-1) - \delta(t-3)\} = \int_0^\infty \; e^{-st}(\delta(t-1) - \delta(t-3))\; dt.$$

To evaluate this integral we use the linearity of the transform and the fact that $\int_b^c \; f(t)\delta(t-a)\; dt$ is equal to $f(a)$ if $a \in (b,c)$ and is equal to zero if $a \notin (b,c)$. Therefore

$$L\{\delta(t-1) - \delta(t-3)\} = \int_0^\infty \; e^{-st}\delta(t-1)\; dt - \int_0^\infty \; e^{-st}\delta(t-3)\; dt$$

$$L\{\delta(t-1) - \delta(t-1)\} = e^{-s(1)} - e^{-s(3)}$$

$$L\{\delta(t-1) - \delta(t-1)\} = e^{-s} - e^{-3s}.$$

9. $t\delta(t-1)$

By definition, the Laplace transform of this function is given by

$$L\{t\delta(t-1)\} = \int_0^\infty \; e^{-st}(t\delta(t-1))\; dt.$$

To evaluate this integral we use the fact that $\int_b^c \; f(t)\delta(t-a)\; dt$ is equal to $f(a)$ if $a \in (b,c)$ and is equal to zero if $a \notin (b,c)$. Therefore

$$L\{t\delta(t-1)\} = \int_0^\infty \; (te^{-st})\delta(t-1)\; dt$$

$$L\{t\delta(t-1)\} = (1)e^{-s(1)}$$

$$L\{t\delta(t-1)\} = e^{-s}.$$

11. $\delta(t-\pi)\sin t$

By definition, the Laplace transform of this function is given by

$$L\{\delta(t-\pi)\sin t\} = \int_0^\infty \; e^{-st}(\sin t\delta(t-\pi)) \; dt.$$

To evaluate this integral we use the fact that $\int_b^c \; f(t)\delta(t-a) \; dt$ is equal to $f(a)$ if $a \in (b,c)$ and is equal to zero if $a \notin (b,c)$. Therefore

$$L\{\delta(t-\pi)\sin t\} = \int_0^\infty \; (\sin t \; e^{-st})\delta(t-\pi) \; dt$$

$$L\{\delta(t-\pi)\sin t\} = \sin(\pi) \; e^{-s(\pi)} = 0.$$

13. $w'' + w = \delta(t-\pi), \; w(0) = 0, \; w'(0) = 0$

Taking transforms of both sides of the equation yields

$$s^2W(s) - s(0) - 0 + W(s) = e^{-\pi s}$$

$$(s^2+1)W(s) = e^{-\pi s}$$

$$W(s) = \frac{e^{-\pi s}}{s^2+1}.$$

Taking inverse transforms gives

$$w(t) = \sin(t+\pi) \; u(t-\pi)$$

$$w(t) = -\sin t \; u(t-\pi).$$

15. $y'' + 2y' - 3y = \delta(t-1) - \delta(t-2), \; y(0) = 2, \; y'(0) = -2$

Taking transforms of both sides of the equation yields

$$s^2Y(s) - s(2) - (-2) + 2[sW(s) - 2] - 3Y(s) = e^{-s} - e^{-2s}$$

$$(s^2 + 2s - 3)Y(s) - 2s - 2 = e^{-s} - e^{-2s}$$

$$(s+3)(s-1)Y(s) = e^{-s} - e^{-2s} + 2s + 2$$

$$(s+3)(s-1)Y(s) = \frac{e^{-s}}{(s+3)(s-1)} - \frac{e^{-2s}}{(s+3)(s-1)} + \frac{2s+2}{(s+3)(s-1)}.$$

Before finding inverse transforms we need to perform two partial fraction expansions. For the first expansion

$$\frac{1}{(s+3)(s-1)} = \frac{A}{s+3} + \frac{B}{s-1}.$$

Finding common denominators and equating numerators gives

$$1 = A(s-1) + B(s+3).$$

Evaluating this equation at appropriate s values we find

$$s = 1: \quad 1 = 4B \longrightarrow B = 1/4$$

$$s = -3: \quad 1 = -4A \longrightarrow A = -1/4$$

so that

$$\frac{1}{(s+3)(s-1)} = -\frac{1}{4(s+3)} + \frac{1}{4(s-1)}.$$

For the second expansion we have

$$\frac{2s+2}{(s+3)(s-1)} = \frac{A}{s+3} + \frac{B}{s-1}.$$

Finding common denominators and equating numerators gives

$$2s + 2 = A(s-1) + B(s+3).$$

Evaluating this equation at appropriate s values we find

$$s = 1: \quad 4 = 4B \longrightarrow B = 1$$

$$s = -3: \quad -4 = -4A \longrightarrow A = 1$$

so that

$$\frac{2s+2}{(s+3)(s-1)} = \frac{1}{s+3} + \frac{1}{s-1}.$$

Combining the partial fraction expansions we find

$$Y(s) = e^{-s}\Big(-\frac{1}{4(s+3)} + \frac{1}{4(s-1)}\Big) - e^{-2s}\Big(-\frac{1}{4(s+3)} + \frac{1}{4(s-1)}\Big) + \frac{1}{s+3} + \frac{1}{s-1}$$

so that taking inverse transforms we find

$$y(t) = (1/4)u(t-1)[-e^{-3(t-1)} + e^{t-1}] - (1/4)u(t-2)[-e^{-3(t-2)} + e^{t-2}] + e^{-3t} + e^{t}.$$

17. $y'' - y = 4\delta(t-2) + t^2,\ y(0) = 0,\ y'(0) = 2$

Taking transforms of both sides of the equation yields

$$s^2Y(s) - s(0) - (2) + Y(s) = 4e^{-2s} + \frac{2}{s^3}$$

$$(s^2-1)Y(s) - 2 = 4e^{-2s} + \frac{2}{s^3}$$

$$(s^2-1)Y(s) = 4e^{-2s} + \frac{2}{s^3} + 2$$

$$(s-1)(s+1)Y(s) = 4e^{-2s} + \frac{2+2s^3}{s^3}$$

$$Y(s) = \frac{4e^{-2s}}{(s-1)(s+1)} + \frac{2+2s^3}{s^3(s-1)(s+1)}$$

Before finding inverse transforms we need to perform two partial fraction expansions. For the first expansion

$$\frac{2+2s^3}{s^3(s-1)(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s+1} + \frac{E}{s-1}.$$

Finding common denominators and equating numerators gives

$$2s^3 + 2 = As^2(s-1)(s+1) + Bs(s-1)(s+1) + C(s-1)(s+1) + Ds^3(s-1) + Es^3(s+1)$$

$$2s^3 + 2 = As^4 - As^2 + Bs^3 - Bs + Cs^2 - C + Ds^4 - Ds^3 + Es^4 + Es^3.$$

Equating like powers of s yields

$$s^0: \quad 2 = -C \quad \longrightarrow \quad C = -2$$

$$s^1: \quad 0 = -B \quad \longrightarrow \quad B = 0$$

$$s^2: \quad 0 = -A + C \quad \longrightarrow \quad A = -2$$

$$s^3: \quad 2 = B - D + E = -D + E$$

$$s^4: \quad 0 = A + D + E = -2 + D + E \quad \longrightarrow \quad D = 0, E = 2$$

so that

$$\frac{2 + 2s^3}{s^3(s-1)(s+1)} = -\frac{2}{s} - \frac{2}{s^3} + \frac{2}{s-1}.$$

For the second expansion we have

$$\frac{4}{(s-1)(s+1)} = \frac{A}{s-1} + \frac{B}{s+1}$$

Finding common denominators and equating numerators gives

$$4 = A(s+1) + B(s-1).$$

Evaluating the equation at appropriate values of s gives

$$s = 1: \quad 4 = 2A \quad \longrightarrow \quad A = 2$$

$$s = -1: \quad 4 = -2B \quad \longrightarrow \quad B = -2$$

so that

$$\frac{4}{(s-1)(s+1)} = \frac{2}{s-1} - \frac{2}{s+1}.$$

Hence we find

$$Y(s) = e^{-2s}\left(\frac{2}{s-1} - \frac{2}{s+1}\right) + -\frac{2}{s} - \frac{2}{s^3} + \frac{2}{s-1}$$

Taking inverse transforms yields

$$y(t) = 2u(t-2)[e^{t-2} - e^{-(t-2)}] - 2 - t^2 + 2e^t.$$

19. $w'' + 6w' + 5w = e^t\delta(t-1),\ w(0) = 0,\ w'(0) = 4$

Taking transforms of both sides of the equation yields

$$s^2W(s) - s(0) - (4) + 6(sW(s) - (0)) + 5W(s) = e^{-s}e$$

$$(s^2 + 6s + 5)W(s) - 4 = e^{-s}e$$

$$(s+1)(s+5)W(s) = e^{-s}e + 4$$

$$W(s) = \frac{e^{-s}e}{(s+1)(s+5)} + \frac{4}{(s+1)(s+5)}.$$

Before finding inverse transforms we need to perform the following partial fraction expansion:

$$\frac{1}{(s+1)(s+5)} = \frac{A}{s+1} + \frac{B}{s+5}.$$

Finding common denominators and equating numerators gives

$$1 = A(s+5) + B(s+1).$$

Evaluating the function at appropriate values of s gives

$$s = -1: \quad 1 = 4A \quad \longrightarrow \quad A = 1/4$$

$$s = -5: \quad 1 = -4B \quad \longrightarrow \quad B = -1/4$$

so that

$$\frac{1}{(s+1)(s+5)} = \frac{1}{4(s+1)} - \frac{1}{4(s+5)}.$$

Hence we find

$$W(s) = ee^{-s}\left(\frac{1}{4(s+1)} - \frac{1}{4(s+5)}\right) + \frac{1}{s+1} - \frac{1}{s+5}.$$

Taking inverse transforms yields

$$w(t) = (1/4)eu(t-1)[e^{-(t-1)} - e^{-5(t-1)}] + e^{-t} - e^{-5t}.$$

21. $y'' + y = \delta(t - 2\pi),\ y(0) = 0,\ y'(0) = 1$

Taking transforms of both sides of the equation yields

$$s^2Y(s) - s(0) - (1) + Y(s) = e^{-2\pi s}$$

$$(s^2+1)Y(s) - 1 = e^{-2\pi s}$$

$$(s^2+1)Y(s) = e^{-2\pi s} + 1$$

$$Y(s) = \frac{e^{-2\pi s}}{s^2+1} + \frac{1}{s^2+1}.$$

Taking inverse transforms yields

$$y(t) = u(t-2\pi)\sin(t-2\pi) + \sin t$$

$$y(t) = u(t-2\pi)\sin t + \sin t.$$

23. $y'' + y = -\delta(t - \pi) + \delta(t - 2\pi),\ y(0) = 0,\ y'(0) = 1$

Taking transforms of both sides of the equation yields

$$s^2Y(s) - s(0) - (1) + Y(s) = -e^{-\pi s} + e^{-2\pi s}$$

$$(s^2+1)Y(s) - 1 = -e^{-\pi s} + e^{-2\pi s}$$

$$(s^2+1)Y(s) = -e^{-\pi s} + e^{-2\pi s} + 1$$

$$Y(s) = -\frac{e^{-\pi s}}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1} + \frac{1}{s^2+1}.$$

Taking inverse transforms yields

$$y(t) = -u(t-\pi)\sin(t-\pi) + u(t-2\pi)\sin(t-2\pi) + \sin t$$

$$y(t) = u(t-\pi)\sin t + u(t-2\pi)\sin t + \sin t.$$